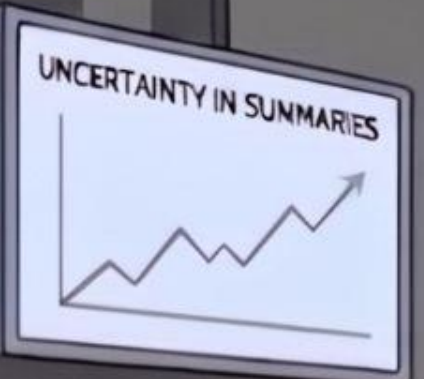


Open Energy Knowledge Gateway

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Energieschweiz & VirtusLab Team





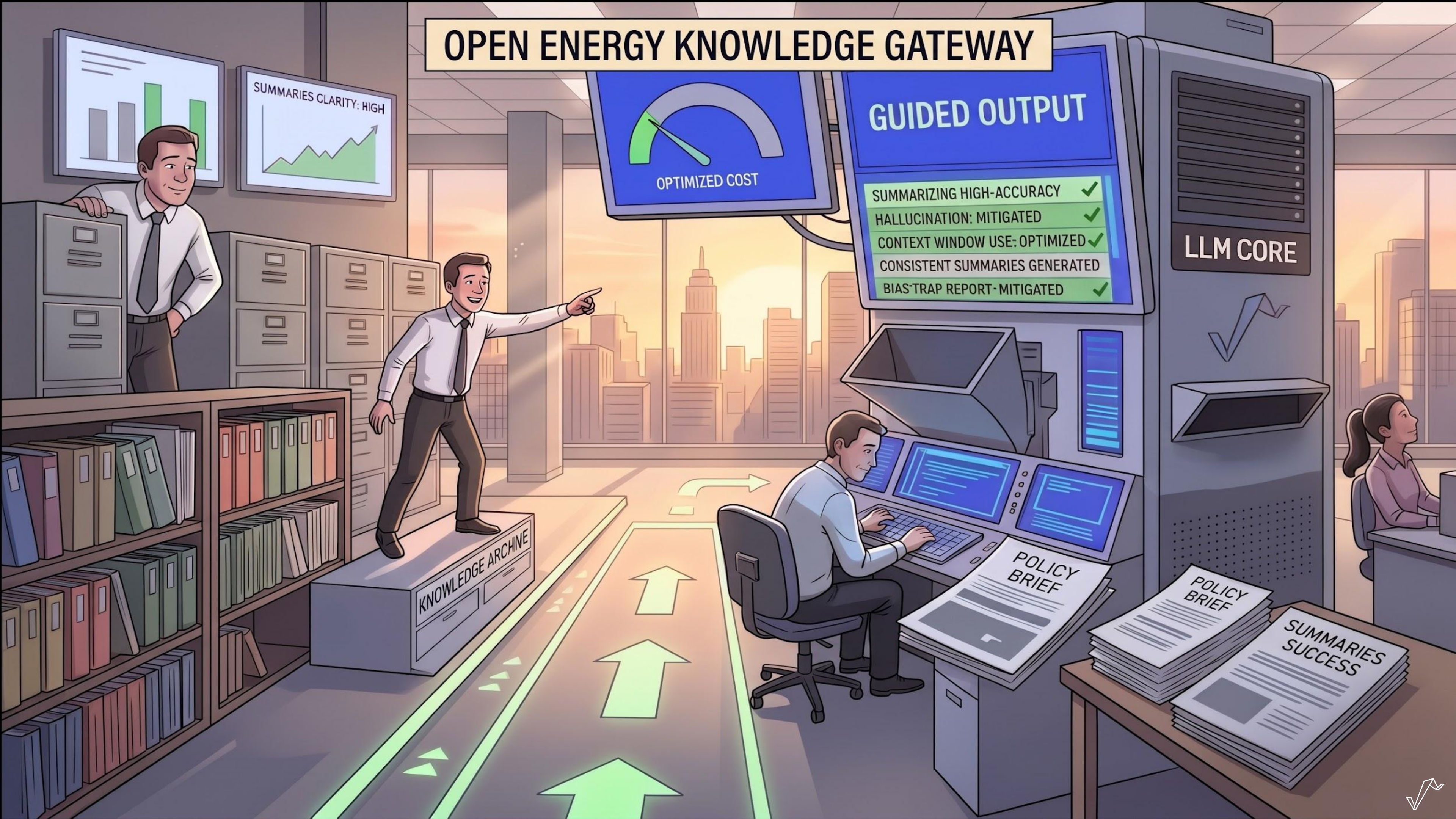
PROCESSING...
SUMMARIZING...
ERROR: HIGH CONTEXT WINDOW USAGE
HALLUCINATION INDEX: **HIGH**
INCONSISTENT SUMMARIES GENERATED

LLM CORE

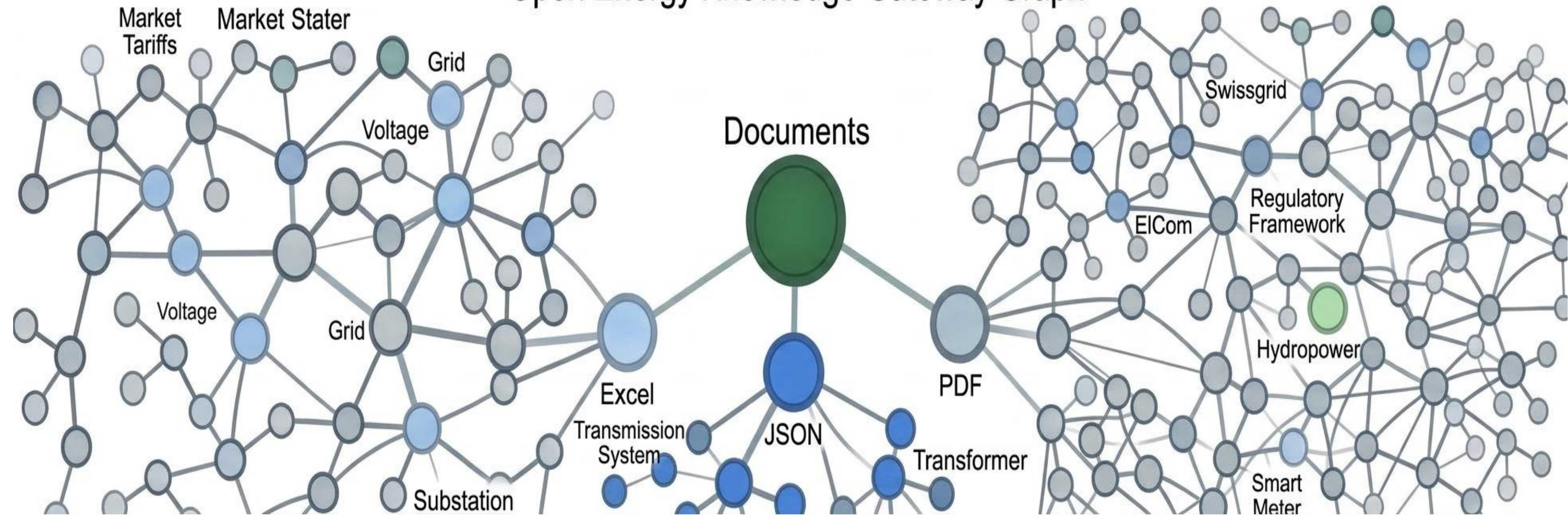
POLICY BRIEF



OPEN ENERGY KNOWLEDGE GATEWAY



Open Energy Knowledge Gateway Graph



Four ways to improve the graph

every correction is one of these four — and each one is a line in a file, not a hand edit of the graph

1

ALIAS

The same thing under two names.

photovoltaics

0 facts

alias

Photovoltaik

50 facts · 15 documents

The question is English, the corpus is German. Nothing bridged them.

2

ADD_FACT

A fact the extraction missed.

Hydropower

2035 target

37,900 GWh

new

It is in the document, it never made it into the graph.

3

REMOVE_FACT

Prose that was never a fact.

Coffee

processed using

Hydropower

Historical prose about old mills surfaced as curated knowledge. 21 such facts hung off one entity.

4

MERGE_ENTITIES

Two nodes, one organisation.

Swiss Federal Office of Energy

merge

SFOE

Both names still resolve — they now point at the same node.

...and how they look in one graph

green was added, salmon is on its way out, dashed means a name that now resolves to a node it is not

Swiss Federal Office of Energy

merge

SFOE

monitors

Energy Strategy 2050

Switzerland

depends on

Hydropower

contributes to

Renewable electricity

hydro power

alias

Hydropower

Coffee

processed using

Hydropower

Hydropower

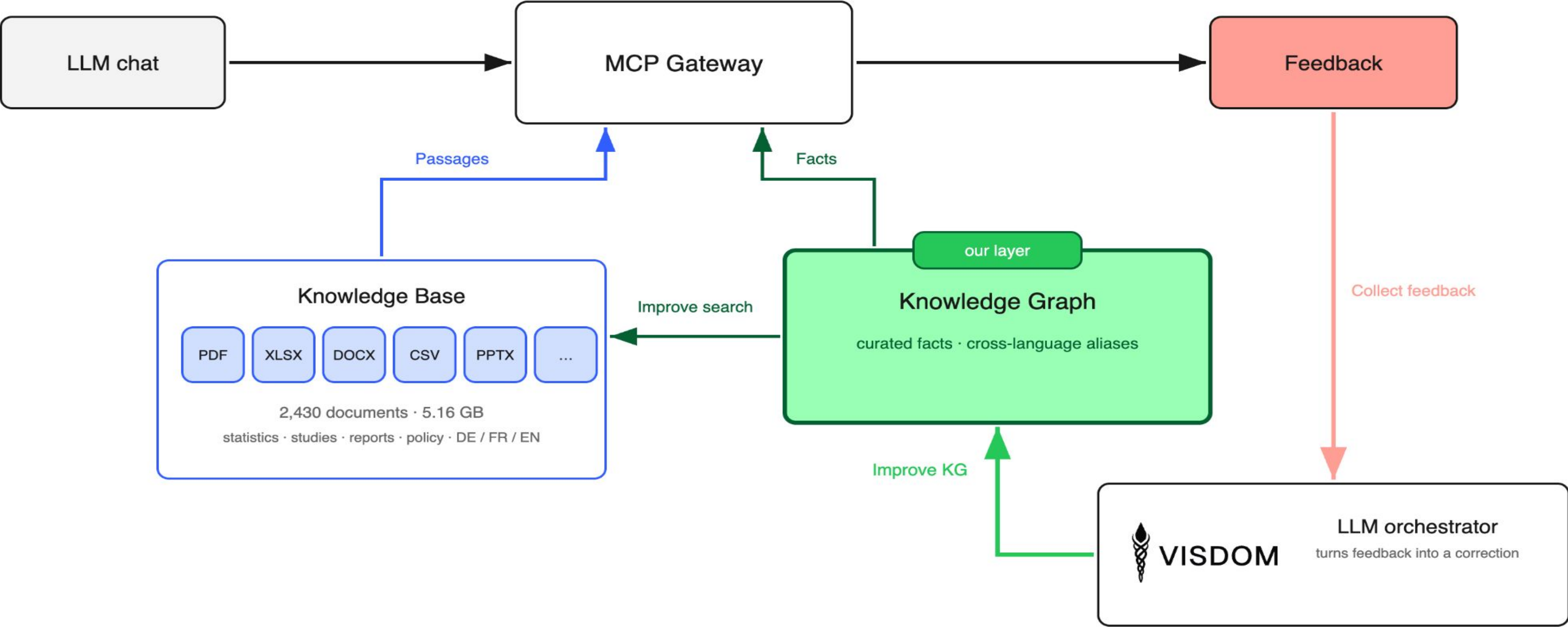
2035 target

37,900 GWh

Nobody edits the graph by hand. A correction is four fields in a file — operation, reason, source, date — and a deterministic applier writes it in.

One answer, two sources

the graph improves the search — the loop improves the graph

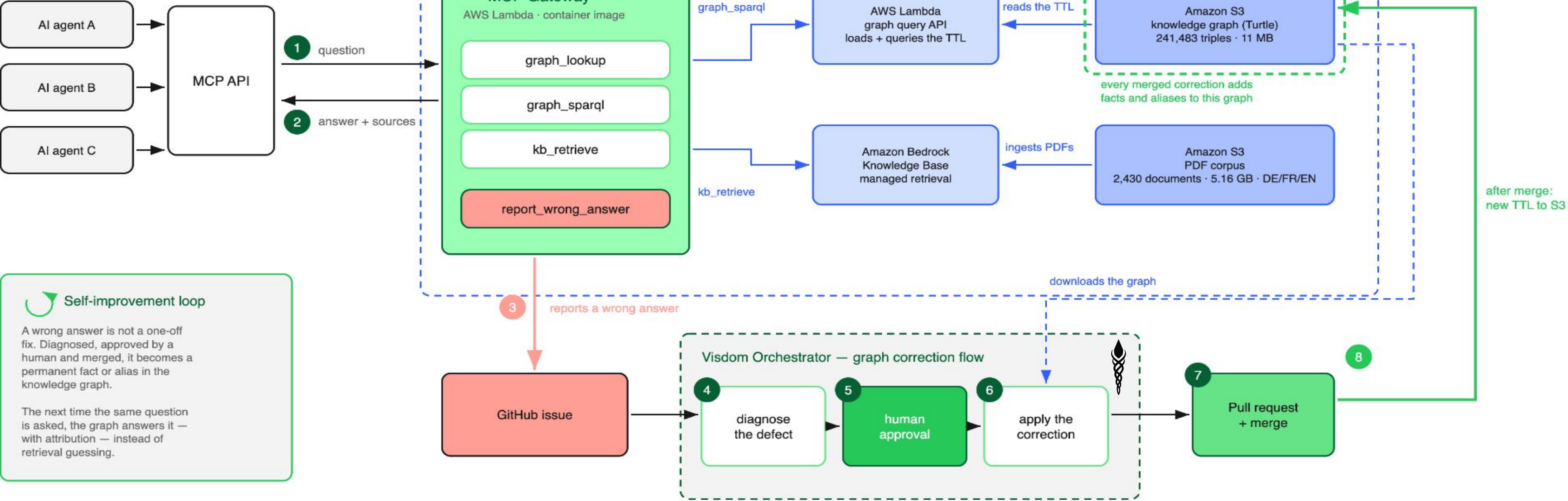


The documents say what is written. The graph says what is true — and the loop keeps it true.

Open Energy Knowledge Gateway

AWS architecture and the human-gated feedback loop: wrong answer → corrected graph

MCP clients · LLM applications



Self-improvement loop

A wrong answer is not a one-off fix. Diagnosed, approved by a human and merged, it becomes a permanent fact or alias in the knowledge graph.

The next time the same question is asked, the graph answers it — with attribution — instead of retrieval guessing.

Legend

- request / control flow
- AWS data path — graph API and knowledge base
- the correction step downloads the graph from Amazon S3
- after merge: the corrected graph is published back to Amazon S3
- a wrong answer is reported straight from the client

Measured live run

diagnose 30 s · human approval · apply 1 m 48 s · validation gate 1.6 s
241,483 → 241,486 triples (+3: one cross-language alias added)
result: a pull request carrying the full graph, changelog, operations file and eval case
the graph on Amazon S3 stays untouched until that pull request is merged

wrong answer → issue → diagnosis → human approval → graph correction → pull request → merge → improved graph answers the next question



Tool calls in real time

- “What are Switzerland's renewable electricity targets?”

aws

Search

[Option+S]

Amazon Bedrock AgentCore > Harness > Harness playground

Harness

Endpoint

Session ID

vl_harness_public_bfe

DEFAULT

cab7d0cc-fae8-42a3-afe7-c

New session

Input: --- | Output: --- | Total: --- | Latency: ---

Now I'll expand the graph by calling the neighbors tool for each extracted label. </think>

Vl-Graph-Neighbors-Tool Vl-Graph-Neighbors-Tool

{ "statusCode": 200, "body": { "entity": "Switzerland", "found": true, "nodes": 1, "direction": "both", "neighb... } }

Vl-Graph-Neighbors-Tool Vl-Graph-Neighbors-Tool

{ "statusCode": 200, "body": { "entity": "renewable targets", "found": false, "nodes": 0, "direction": "both", "... } }

1.0s

Let me try a shorter form for renewable targets: "renewable" and "targets" separately. </think>

+ Show more (+5)

Now let me retrieve information about Switzerland's renewable electricity targets. I'll query the knowledge base using relevant terms. </think>

Vl-KB-Bfe-Public Retrieve

{ "retrievalResults": [{ "content": { "text": "This arrow corresponds to a single but significant goal: - Electricity from new... } }] }

1.8s

Based on the retrieved information from Switzerland's Energy Strategy 2050 documents, I can provide the renewable electricity targets:

According to Switzerland's Energy Strategy 2050, the long-term goal is to generate all electricity from renewable sources.

Write a prompt (Enter = send, Shift+Enter = new line, Arrow Up/Down = history, Esc = clear)

Graph lookup tool call

VL-Graph-Neighbors-Tool VL-Graph-Neighbors-Tool Success

Input

```
√ { 2 items
  "entity": "Switzerland"
  "direction": "both"
}
```

Output

```
√ { 2 items
  "statusCode": 200
  "body": [{"entity": "Switzerland", "found": true, "nodes": 1, "direction": "both", "label": "105'927 fans", "via": [{"predicate": "sold", "direction": "out"}]}, {"label": "23.1 TWh", "direction": "out"}]}, {"label": "3 Gigawatt elektrisch", "via": [{"predicate": "had a to 36 % of total electricity production", "via": [{"predicate": "had", "direction": "out"}]}, {"label": "48'545 water", "direction": "out"}]}, {"label": "93'171 electric motors", "via": [{"predicate": "has nuclear power plants", "direction": "out"}]}]}
```

Retrieve tool call

VI-KB-Bfe-Public Retrieve Success

Input

```
√ { 1 item
  √ "retrievalQuery": { 1 item
    √ "text": "Switzerland renewable electricity targets Energy Strategy 2050"
  }
}
```

Output

```
√ { 1 item
  √ "retrievalResults": [ 5 items
    √ 0: { 5 items
      √ "content": { 2 items
        √ "text": "This arrow corresponds to a si..."
        √ "type": "TEXT"
      }
      √ "documentId": "s3://sandbox-bfe-public-data-p..."
      √ "location": { 2 items
        √ "s3Location": { 1 item
          √ "uri": "https://sandbox-bfe-public-dat..."
        }
        √ "type": "S3"
      }
    }
    √ "metadata": { 10 items
      √ "_file_type": "PDF"
      √ "_document_title": "2021-08-04_fakten-zur-energie-..."
      √ "_source_uri": "https://sandbox-bfe-public-dat-..."
    }
  ]
}
```

Tool calls in real time

“What are Switzerland's renewable electricity targets?”

aws

Search

[Option+S]

Ask Amazon Q

Amazon Bedrock AgentCore

Harness

Harness playground

Harness

Endpoint

Session ID

vl_harness_public_bfe

DEFAULT

cab7d0cc-fae8-42a3-afe7-c

New session

Input: ---

Output: ---

Total: ---

Latency: ---

added to the relevant data source.

I should use the vl-feedback tool to record this feedback. I need to provide:

- user_question: the original question about Switzerland's renewable electricity targets
- answer: my previous answer about the 14.5 TWh target
- user_reaction: the user's feedback that reducing energy import should be added as a goal
- source: I don't have specific session information, so I'll omit this

Let me format this properly. </think>

VL-Feedback-Tool VL-Feedback

```
{\"statusCode\":200,\"body\":{\"status\": \"accepted\", \"bucket\": \"vl-bfe-kg-bfe-artifacts\", \"key\": \"feedback-2026-0...\"}}
```

I've logged your feedback that reducing energy import should be added as one of Switzerland's renewable electricity goals. The feedback has been recorded in the system for future reference and potential inclusion in the knowledge base.

Is there anything else you'd like to know about Switzerland's renewable energy targets or policies?

13:37:05 CEST

Write a prompt (Enter = send, Shift+Enter = new line, Arrow Up/Down = history, Esc = clear)

Feedback tool call

cg-bfe-artifacts

vl-bfe-kg-bfe-artifacts

Objects

Metadata

Properties

Permissions

Metrics

Objects (3)

Copy S3 URI

Copy URL

Download

Open

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inv](#) need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Show versions

☐	Name	Type
☐	feedback-2026-09-10.md	md
☐	feedback-2026-09-11.md	md



Governance & Operating Model

— Six recommendations — and what each one rests on

1 · Ownership

The office owns the documents. One editor owns the graph. Anyone proposes, nobody commits. Every fact carries reason, source, author, date.

editorial, not infrastructure

2 · Freshness

Update on publication, not on a calendar. last_updated per document, graph_version per answer.

stable document ids are mandatory

3 · Scope

Only what is already published and citable. Deduplicate before exposure. Fix language metadata at the source.

305 of 2,430 objects are duplicates – 12.6%

4 · Access

Public to read. Authenticated to write or to spend. Anonymous reads are rate-limited; named clients get higher quotas.

feedback is a write – it must be attributable

5 · Rollout & cost

Prove it internally before promising anything. Internal → named pilots → public with a published p95.

14 s cold / 3.6 s warm – too early for SLA

6 · Consumer duties

We supply facts with sources. The application owns the answer. Always cite. Treat "not covered" as an answer. Report wrong answers back.

61 numbers checked · 0 hallucinated · 6 composition defects



Let's Talk During Apéro



Kseniia Sych



Tomasz Jażdżewski



Paweł Urba



Grzegorz Król